MATH-540. HW#3.

Lazizbek.

Problem 1. (Z&T, Ch 1, Pr. 3.4)

Suppose the curve, C, is given parametrically by

(1) $$C: (x, y, z) = (e^t \cdot \cos(t),\ e^t \sin(t),\ e^{2t})$$

(i) Find the level surface, S_c, of

(2) $$S_c: F(x, y, z) = 2(x^2 + y^2 + 3) - 2z$$

upon which the curve, C, lies. (ii) Verify that at each pt of C, the tangent vector to C is $\perp$ to the vector normal to S_c.

Sol'n:

(i) Find the level Surface S_c:

C lies on the level surface, S_c, means

(3) $$S_c: F(x, y, z) = D \text{ (for some constant } D) \ \&$$

$C \in S_c$, so

(4) $$F(x, y, z)\Big|_C = D \implies$$

$$\left[2(x^2 + y^2 + 3) - 2z\right]\Big|_C = 2\left[(e^t \cos(t))^2 + (e^t \cdot \sin(t))^2 + 3\right] - 2e^{2t} = D$$

$$\Rightarrow 2 \cdot \left[e^{2t}\left(\cos^2(t) + \sin^2(t)\right) + 3\right] - 2e^{2t} = D$$

$$2e^{2t} + 6 - \cancel{2e^{2t}} = D$$

(5) $$\boxed{D = 6}, \quad \text{so } S_c \text{ is}$$

$$S_c: F(x, y, z) = 6 \implies$$

$$2(x^2 + y^2 + 3) - 2z = 6 \quad \text{or}$$

(6) $$\boxed{z = x^2 + y^2}.$$

(ii) Verify that @ each pt of C, the tangent vector to C is $\perp$ to the normal vector to S_c:

- Tangent to C is derivative of function of C.
- Normal to S_c is gradient of function of S_c

Recall that by (1), curve C is given parly by

(1) $C: \quad (x, y, z) = (e^t \cos(t),\ e^t \sin(t),\ e^{2t})$

so tangent vector, $\underline{T}_c(t) = \langle x'(t),\ y'(t),\ z'(t) \rangle$

(7) $\underline{T}_c(t) = \langle (e^t \cos(t) - e^t \sin(t)),\ (e^t \sin(t) + e^t \cos(t)),\ 2e^{2t} \rangle$

Also by (6), surface S_c is given by a general for-a:

(6) $z = x^2 + y^2$, so $\boxed{x^2 + y^2 - z} = 0$, we need to find the gradient of this:

(8) $\underline{\nabla} F(x, y, z) = \langle 2x,\ 2y,\ -1 \rangle$.

Now to verify $\underline{T}_c(t) \perp \underline{\nabla} F(x, y, z)$, we need to turn $\underline{\nabla} F(x, y, z)$ into a function of (t). Remember $\nabla F(x, y, z)$ is a gradient, which is a normal vector to the surface $F(x, y, z) = 6$ at each (x, y, z) pt, so the gradient on the curve, $C = (x(t),\ y(t),\ z(t))$, is evaluated simply by restricting $\underline{\nabla} F(x, y, z)$ onto the curve C (i.e. by replacing $\underline{\nabla} F(x, y, z)$ by $\underline{\nabla} F(x(t),\ y(t),\ z(t))$).

Easy way to see $\underline{T}_c(t) \perp \underline{\nabla} F(x, y, z)\big|_c$ is calculating dot pr.:

(9) $\langle [e^t \cos(t) - e^t \sin(t)],\ [e^t \sin(t) + e^t \cos(t)],\ 2e^{2t} \rangle \cdot \langle 2e^t \cos(t),\ 2e^t \sin(t),\ -1 \rangle$

$= 2e^{2t} \cdot [\cos^2(t) - \cancel{\cos(t) \cdot \sin(t)} + \sin^2(t) + \cancel{\sin(t) \cos(t)}] - 2e^{2t} = 2e^{2t} - 2e^{2t} = 0$ ✓

So $\underline{T}_c(t)$ is orthogonal to $\underline{\nabla} F(x(t),\ y(t),\ z(t))$ as desired. ∎